Weekly Diary – Week 17

Internship Organization: Vikaspedia
Duration: 28/04/2026 – 03/05/2026

Objective of the Week

The objective of this week was to improve the efficiency of the crawling engine by implementing parallel processing, batch handling, redirect validation, and error management mechanisms required for scanning large numbers of URLs efficiently.

Date- 28/April

Focused on improving the scalability of the crawler so that it could process large batches of URLs efficiently. Studied different approaches for implementing concurrent request handling using multi-threading and thread pools. Analyzed how processing URLs sequentially can significantly reduce performance when dealing with datasets containing thousands of links. Began implementing batch-based processing logic where groups of URLs are processed together instead of individually. Conducted small performance tests to compare sequential and parallel execution speeds.

Date- 29/April

Implemented multi-threading functionality in the crawler using controlled thread pools for concurrent URL validation. Configured the system to process multiple links simultaneously while monitoring memory and CPU usage. Worked on balancing thread execution to prevent system overload and unstable request handling. Conducted testing with sample batches of URLs and observed significant improvements in crawling speed. Also analyzed how uncontrolled concurrency may lead to request failures or server blocking.

Day- 30/April

Developed rate-limiting functionality to prevent the crawler from sending too many requests to the same server within a short duration. Implemented retry mechanisms with delay intervals to handle temporary network failures and server-side errors. Added timeout configuration to stop excessively slow requests from blocking the crawling process. Conducted tests with intentionally slow or unreachable URLs to verify that the crawler handled failures gracefully without interrupting overall execution.

Date- 1/May

Improved redirect-handling functionality by implementing logic to follow multiple redirect chains automatically. Added validation mechanisms to ensure that redirected URLs eventually resolve to a working destination. Implemented redirect loop detection to identify URLs that continuously redirect between the same locations. Tested the crawler with redirected, shortened, and invalid URLs to verify accurate redirect processing and classification.

Date- 2/May

Enhanced the logging and reporting functionality of the crawler system. Implemented detailed logs for successful requests, failed URLs, timeout errors, redirects, and retry attempts. Structured the logs in a readable format so that crawler behavior could be analyzed easily during debugging and performance testing. Also added counters and summary statistics for processed URLs, successful validations, and detected failures.

Date- 3/May

Conducted large-scale testing of the crawler with a mixed dataset containing valid links, broken URLs, redirected pages, and timeout-prone domains. Monitored system performance during concurrent execution and identified areas where optimization was required. Improved exception handling to ensure uninterrupted processing during unexpected failures. Documented observations related to scalability, crawler stability, and request-handling efficiency during high-volume processing.

Conclusion

By the end of this week, the crawler system was significantly improved with support for multi-threaded execution, batch processing, rate limiting, retry handling, and redirect validation. The system demonstrated better scalability and performance while processing large numbers of URLs concurrently.